Computer Practical

Gaussian Plume Model

Atmospheric dispersion from point sources

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1 Overview

In this practical you will use a Gaussian plume model to investigate how pollutants emitted from an elevated point source are dispersed by the atmosphere. You will change one assumption at a time—wind direction, atmospheric stability, number of sources and aerosol water uptake—and interpret the resulting concentration fields.

The model is deliberately simple. It is designed to expose the physical assumptions rather than to reproduce a full regulatory air-quality model. Keep a practical record: save each figure using a meaningful filename and note the settings used for the run.

By the end of the practical you should be able to:

- explain the main assumptions behind a steady Gaussian plume model;
- relate wind direction and atmospheric stability to downwind pollutant concentrations;
- interpret the Pasquill–Gifford stability classes A–F and their effect on plume spread;
- explain how contributions from several point sources add together;
- distinguish dry particulate mass from an idealised humidified mass loading;
- recognise important limitations of a simple Gaussian plume calculation.

2 From advection–diffusion to a Gaussian plume

For a pollutant concentration C, the three-dimensional advection–diffusion equation can be written schematically as

$$\frac{\partial C}{\partial t} + u\frac{\partial C}{\partial x} + v\frac{\partial C}{\partial y} + w\frac{\partial C}{\partial z} = K_x\frac{\partial^2 C}{\partial x^2} + K_y\frac{\partial^2 C}{\partial y^2} + K_z\frac{\partial^2 C}{\partial z^2}. \tag{1}$$

For the teaching model we assume a steady plume, a constant mean wind along the downwind direction, negligible mean crosswind and vertical advection, and negligible diffusion along the wind compared with advection. This leaves

$$u\frac{\partial C}{\partial x} = K_y\frac{\partial^2 C}{\partial y^2} + K_z\frac{\partial^2 C}{\partial z^2}. \tag{2}$$

For a continuous elevated point source of strength Q, one solution has the familiar Gaussian form

$$C(x,y,z) = \frac{Q}{2\pi u\sigma_y\sigma_z}\exp\left(-\frac{y^2}{2\sigma_y^2}\right)\left[\exp\left(-\frac{(z-H)^2}{2\sigma_z^2}\right) + \exp\left(-\frac{(z+H)^2}{2\sigma_z^2}\right)\right]. \tag{3}$$

Here H is stack height and σ_y and σ_z describe the lateral and vertical spread of the plume. The second vertical exponential is an “image source” that represents reflection at the ground.

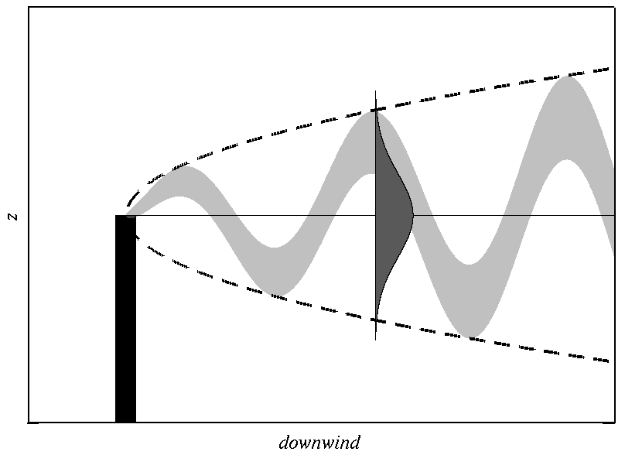

Figure 1: Schematic of the scenario being modelled, where H is the height of the stack.

Stability matters through σ_y and σ_z. The code uses empirical Pasquill–Gifford/ISC-style dispersion fits. Class A is very unstable and class F very stable. Unstable air generally produces faster plume spread; stable air suppresses vertical mixing and can keep a plume narrow for longer. The consequence at one receptor depends on both mixing and distance from the source.

3 Log in and download the code

Use the same SSH/SFTP workflow as in *Tools of the Trade*. On campus, the teaching-server address is `130.88.55.57`; otherwise use the server details supplied for the course. Replace `YOUR_USERNAME` with your own username.

In your SSH window:

```
ssh -Y YOUR_USERNAME@130.88.55.57
```

Open a second terminal window for file transfer:

```
sftp YOUR_USERNAME@130.88.55.57
```

Your password will not be displayed while you type it.

File-transfer reminder. Keep the SSH window open for editing and running the model, and use the second SFTP window for retrieving your results. Commands such as `python3`, `nano` and `git` are entered in the **SSH window**; `get` is entered in the **SFTP window**.

In SFTP, the general form is:

```
get REMOTE_FILE LOCAL_FILENAME
```

The first filename is the file on the server. The optional second filename is the name given to the downloaded copy on your own computer. Use a different, meaningful local filename for each experiment so that later runs do not overwrite your earlier results.

In the SSH window, obtain the repository and enter the Python directory:

```
git clone https://github.com/EnvModelling/gaussian-plume-model-practical
cd gaussian-plume-model-practical/python
pwd
ls
```

You should see `gaussian_plume_model.py`, `gauss_func.py`, `calc_sigmas.py`, `overlay_on_map.py` and `map_green_lane.mat`.

4 Inspecting and running the model

Open the main script with line numbers displayed:

```
nano -l gaussian_plume_model.py
```

In `nano`, `Ctrl-W` searches. Search for `PARAMETERS TO CHANGE`. In the current repository the editable block is around lines 79–101. Line numbers are approximate; searching for a parameter name is more reliable if the code is updated.

Run the model with:

```
python3 gaussian_plume_model.py
```

The model prints the full path of the output it creates. The default run writes

```
/tmp/YOUR_USERNAME/plan_view.png
```

In the SFTP window, download it and give the local copy a useful name:

```
get /tmp/YOUR_USERNAME/plan_view.png gpm_default.png
```

Here the server file remains `plan_view.png`, while the copy on your own computer is saved as `gpm_default.png`. Use the same pattern throughout the practical, changing the local filename to identify each experiment.

Controlled comparisons. The parameter `RANDOM_SEED = 22001` makes the synthetic wind sequence reproducible. Keep this seed unchanged while comparing stability classes, stack numbers or aerosol assumptions. Otherwise you would change the meteorology at the same time as the parameter you are trying to study.

To restore the supplied version of tracked files when instructed:

```
git reset --hard
```

This discards edits inside the repository. It does not delete figures already saved in `/tmp/$USER/` or downloaded to your computer.

5 Parameters used in the practical

Table 1: Main editable parameters in `gaussian_plume_model.py`.

`Parameter`	Default	Meaning
`RH`	0.90	Relative humidity used by the idealised water-uptake calculation
`AEROSOL_TYPE`	SODIUM_CHLORIDE	Particle composition
`HUMIDIFICATION`	DRY_AEROSOL	Use dry mass or apply idealised water uptake
`STABILITY_CLASS`	1	Pasquill–Gifford class: 1=A through 6=F
`STABILITY_USED`	CONSTANT_STABILITY	Fixed class or idealised annual cycle
`OUTPUT`	PLAN_VIEW	Plan view, vertical slice, receptor time series or map overlay
`X_SLICE`	25	Python grid index 25 corresponds to $x = 0$ m
`Y_SLICE`	0	Python grid index 0 corresponds to $y = -2500$ m
`WIND`	PREVAILING_WIND	Constant, random or prevailing wind direction
`WIND_SPEED`	5.0 m s^{-1}	Wind speed; held fixed in these experiments
`RANDOM_SEED`	22001	Seed for reproducible synthetic wind directions
`STACKS`	ONE_STACK	Number of active stacks (1–3)
`Q`	40 g s^{-1} each	Dry pollutant mass emission rate
`H`	50 m each	Stack heights
`DAYS`	50	Number of 24-hour periods included in the run

Table 2: Pasquill–Gifford classes used by the teaching model.

Value	Class	Description
1	A	very unstable
2	B	moderately unstable
3	C	slightly unstable
4	D	neutral
5	E	moderately stable
6	F	very stable

Table 3: Output modes and remote files.

Setting	What is plotted	Remote output
`PLAN_VIEW`	time-mean surface concentration	`plan_view.png`
`HEIGHT_SLICE`	time-mean y–z concentration slice	`height_slice.png`
`SURFACE_TIME`	concentration at one surface receptor	`surface_time.png`
`OVERLAY`	mean surface contours on the supplied map	`overlay.png`

6 Experiments

For every experiment, record the settings you changed, save the output, and write one or two sentences describing what changed physically. Unless instructed otherwise, leave all unlisted parameters at their supplied defaults.

6.1 Wind direction and variability

First isolate the effect of wind direction. Restore the defaults, then set neutral stability and one stack:

```
STABILITY_CLASS = 4
STABILITY_USED = CONSTANT_STABILITY
STACKS = ONE_STACK
OUTPUT = PLAN_VIEW
```

Run the three cases below. Keep `RANDOM_SEED` fixed.

Run	Set `WIND` to	Suggested local filename
A	`CONSTANT_WIND`	`wind_constant.png`
B	`FLUCTUATING_WIND`	`wind_random.png`
C	`PREVAILING_WIND`	`wind_prevailing.png`

Exercise. Before comparing the plots, predict the shape expected for each wind assumption. Then explain how wind-direction variability changes the area exposed to pollutant and the maximum time-mean concentration. Why is a prevailing wind different from a completely random wind even though both vary from hour to hour?

6.2 Multiple stacks

Keep neutral stability and `WIND = PREVAILING_WIND`. Run one, two and three active stacks:

Run	Set STACKS to	Suggested local filename
A	`ONE_STACK`	`stacks_1.png`
B	`TWO_STACKS`	`stacks_2.png`
C	`THREE_STACKS`	`stacks_3.png`

The source positions, emissions and heights are stored in `STACK_X`, `STACK_Y`, `Q` and `H`. The first `STACKS` entries are used.

Exercise. Where do the concentration fields from individual stacks overlap? In the code, concentrations from separate stacks are simply added. Explain why this superposition is possible for this linear plume model. If time permits, move one stack or change its height and predict the effect before rerunning.

6.3 Atmospheric stability: surface plan view

Restore the defaults, then use one stack, a prevailing wind and a fixed stability class:

```
OUTPUT = PLAN_VIEW
STACKS = ONE_STACK
WIND = PREVAILING_WIND
STABILITY_USED = CONSTANT_STABILITY
```

Run classes A–F by setting `STABILITY_CLASS` from 1 to 6. Download each `plan_view.png` before the next run; for example use `stability_A_plan.png`, . . . , `stability_F_plan.png`.

Exercise. Compare plume width, downwind extent and ground-level concentration as stability changes. Link your explanation to the behaviour of σ_y and σ_z. Is "more unstable" always the same as "higher concentration at every ground receptor"? Explain why not.

6.4 Atmospheric stability: vertical structure

Now set

```
OUTPUT = HEIGHT_SLICE
X_SLICE = 25
```

so that the model plots a y–z slice at $x = 0$ m. Repeat the stability classes A–F, downloading each `height_slice.png` before the next run.

Exercise. How does stability alter the vertical spread of the plume and the fraction of pollutant that reaches the surface? Relate the results qualitatively to the textbook plume regimes often called looping, coning and fanning. Note that this is a steady, time-mean Gaussian model: it changes plume spread but does not resolve individual turbulent "loops" in the instantaneous plume.

6.5 An idealised annual cycle in stability

The model can impose a deliberately simple seasonal cycle: stable classes in winter and unstable classes in summer. It does *not* include a diurnal stability cycle. Restore the defaults, then set

```
OUTPUT = SURFACE_TIME
STABILITY_USED = ANNUAL_CYCLE
X_SLICE = 25
Y_SLICE = 0
DAYS = 365
```

This receptor is at $x = 0$ m, $y = -2500$ m. Run the model and download

```
get /tmp/YOUR_USERNAME/surface_time.png surface_annual.png
```

The upper panel shows hourly concentration plus a 24-hour moving mean; the lower panel shows the imposed stability class.

Exercise. Describe the seasonal pattern at this receptor. Use both distance from the stack and plume spread to explain why the relationship between atmospheric instability and concentration at a particular point is not necessarily monotonic. What is realistic about this experiment, and what is intentionally artificial?

6.6 Idealised aerosol water uptake and map overlay

The original teaching code included a simplified humidity calculation. It is retained here, but it should not be described as a full Köhler model: the Kelvin curvature term and non-ideal solution effects are omitted. Under the ideal-solution approximation used here, the wet/dry particle-mass factor is

$$G_m = \frac{m_{\mathrm{wet}}}{m_{\mathrm{dry}}} = 1 + \frac{RH}{1 - RH}\,\nu\,\frac{M_w}{M_s}, \tag{4}$$

where ν is an idealised van't Hoff factor, M_w is the molar mass of water and M_s is the molar mass of the dry solute. The factor multiplies the plume mass concentration; it does not change the dispersion itself.

Restore the defaults, then set

```
OUTPUT = OVERLAY
STABILITY_CLASS = 4
STABILITY_USED = CONSTANT_STABILITY
WIND = PREVAILING_WIND
DAYS = 50
```

Run the following cases:

Run	HUMIDIFICATION	AEROSOL_TYPE	Local filename
A	DRY_AEROSOL	SODIUM_CHLORIDE	overlay_dry.png
B	HUMIDIFY	SODIUM_CHLORIDE	overlay_nacl_wet.png
C	HUMIDIFY	ORGANIC_ACID	overlay_organic_wet.png

When humidification is enabled, the script prints the wet/dry mass factor used.

Prediction. Using $RH = 0.90$, calculate G_m for sodium chloride ($\nu = 2$, $M_s = 58.44$ g mol^{-1}) and for the idealised organic acid ($\nu = 1$, $M_s = 200$ g mol^{-1}). Then run the model. Do the contours change shape, magnitude, or both? Explain why.

7 What this model leaves out

A useful model is defined as much by its assumptions as by its equations. This practical model does **not** include plume rise from stack temperature or momentum, terrain, buildings and wake effects, deposition, chemical reactions, changing wind speed with height, measured meteorology, or a realistic boundary layer. The source is continuous and the Gaussian equation represents a steady mean plume for each hourly meteorological state.

What you should take away. A Gaussian plume is a compact way to connect emissions, wind and turbulent dispersion to concentrations. The calculation can be very informative, but its predictions are conditional on the assumptions used for u, σ_y, σ_z, source geometry and meteorology. When you interpret a model result, always ask which process was represented, which was parameterised, and which was omitted.